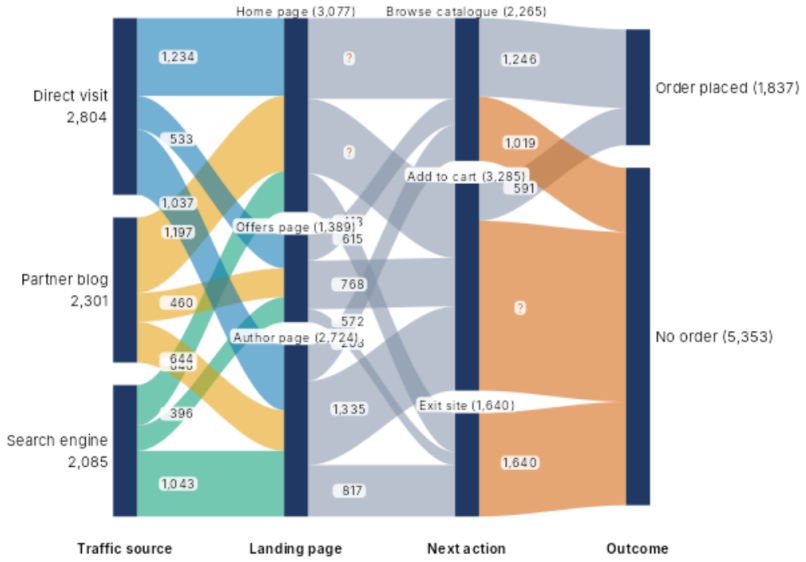
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DSA06 - Data Handling and Visualization | Activity 6: Website Flow Sankey Design | Slot D

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Scenario: The analytics team of an online bookstore tracked 7,190 sessions last month through four stages: traffic source, landing page, next action and outcome. The Sankey diagram below shows the number of sessions on every flow. Three flows are hidden and marked "?".



(a) Using conservation of flow (what enters a node equals what leaves it) find the three hidden flows, showing working: Home page to Browse catalogue; Add to cart to No order; Home page to Add to cart. State the rule you used. [4]

(b) Compute: (i) the overall conversion rate, i.e. Order placed as a percentage of all sessions; (ii) the largest single flow into "No order" or "Exit site"; (iii) the percentage of "Home page" sessions that go on to "Add to cart". [4]

(c) Design the final Sankey: give four design decisions (node order to reduce crossings, colour rule, labels, what to highlight) and justify each in one line using this data. [4]

(d) The Sankey is to be printed with 1 cm of node height = 400 sessions. Compute the height in cm of each landing-page node (Offers page, Author page, Home page). [2]

(e) Write R code: a links data frame for the first four flows leaving "Search engine" and "Partner blog" in the table order of the diagram, a nodes data frame, and the networkD3::sankeyNetwork() call (or plotly) that draws the whole diagram. [4]

(f) State two situations in which a Sankey diagram misleads or becomes unreadable, and name one alternative chart for each. [2]

(a) Find the three hidden flows

Rule used: Conservation of flow — the total number of sessions entering a node equals the total number leaving it.

1. Home page → Browse catalogue

$$2265-(615+1335)=315$$

Answer: 315 sessions

2. Home page → Add to cart

$$3077-(315+615)=2147$$

Answer: 2,147 sessions

These calculations use the visible flow labels as interpreted from the diagram.

3. Add to cart → No order

The Add to cart node has 3,285 sessions, of which 591 lead to an order.

$$3285-591=2694$$

Answer: 2,694 sessions

(b) Calculate the following

(i) Overall conversion rate

$$\text{Conversion Rate}=\frac{\text{Orders placed}}{\text{Total sessions}}\times100$$

$$=\frac{1837}{7190}\times100=\boxed{25.54\%}$$

(ii) Largest single flow into No order or Exit site

The largest single flow is Add to cart → No order = 2,694 sessions.

(iii) Percentage of Home page sessions that go to Add to cart

$$\frac{2300}{3077}\times100=\boxed{74.75\%}$$

(c) Four Sankey diagram design decisions

1. Node order: Arrange nodes in four stages — Traffic source, Landing page, Next action, and Outcome — to show the direction of movement clearly.
2. Colour rule: Use different colours for traffic sources and consistent colours for the same categories to make flows easier to trace.
3. Labels: Display node names, total sessions, and flow values to make the diagram understandable.
4. Highlighting: Highlight the 1,837 orders placed and the 2,694 sessions ending in no order to emphasize conversion and drop-off.

(d) Calculate landing-page node heights

Formula:

$$\text{Height in cm}=\frac{\text{Sessions}}{400}$$

Landing page	Sessions	Height
Offers page	1,389	3.47 cm
Author page	2,724	6.81 cm
Home page	3,077	7.69 cm

(e) R code for the Sankey diagram

```
library(networkD3)


nodes <- data.frame(name = c(
  "Direct visit", "Partner blog", "Search engine",
  "Home page", "Offers page", "Author page",
  "Browse catalogue", "Add to cart", "Exit site",
  "Order placed", "No order"
))


links <- data.frame(
```

```
  source = c(
    0, 0, 0, 1, 1, 1, 2, 2, 2,
    3, 3, 3, 4, 4, 4, 5, 5, 5,
    6, 6, 7, 7, 8
  ),
  target = c(
    3, 4, 5, 3, 4, 5, 3, 4, 5,
    6, 7, 8, 6, 7, 8, 6, 7, 8,
    9, 10, 9, 10, 10
  ),
  value = c(
    1234, 533, 1037, 1197, 460, 644,
    646, 396, 1043,
    162, 2300, 615, 768, 168, 453,
    1335, 817, 572,
    1246, 1019, 591, 2694, 1640
  )
)

sankeyNetwork(
  Links = links,
  Nodes = nodes,
  Source = "source",
  Target = "target",
  Value = "value",
  NodeID = "name",
```

```
  fontSize = 12,
  nodeWidth = 25
)
```

Note: The source and target indices start from 0 in networkD3. The code uses the inferred hidden-flow values calculated above.

(f) Two situations where a Sankey diagram can mislead or become unreadable

1. Too many nodes and crossing flows: The diagram becomes cluttered and difficult to follow. Alternative: Use a table or grouped bar chart.
2. Very small flows alongside very large flows: Small flows may be difficult to see, and viewers may overlook them. Alternative: Use a bar chart with labelled values.